Introduction and Study Objectives

Background

- Diarrhoeal disease remains a major public health challenge in low- and middle-income countries.
- Unsafe drinking water is a major contributor to diarrhoeal morbidity and mortality, with millions of cases reported globally each year [1].
- Understanding the association between water source and diarrhoea is essential for effective prevention strategies.

Main Objective

- To estimate the incidence of diarrhoea and assess its association with household water source.

Specific Objectives

- Estimate diarrhoeal incidence rate using person-time.
- Compare incidence between tap and well water users.
- Assess socio-demographic risk factors.

Methodology and Statistical Analysis

Study Design and Setting

- **Design:** Prospective cohort study (12-month follow-up)
- **Sample:** 1,190 participants (10 excluded due to inconsistency data)
- **Exposure:** Household water source (Tap vs. Well)
- **Outcome:** At least one episode of diarrhoea

Statistical Methods

- **Baseline Comparison:** χ^2 test for categorical variables; Wilcoxon test for continuous variables.
- **Incidence Risk (Ratio):** Estimated using the Binomial distribution .
- **Incidence Rate (Ratio):** Estimated using Poisson regression, accounting for person-time.

Baseline Characteristics by Water Source

Table 1: Baseline characteristics stratified by water source

Variable	Level	Tap ($n = 618$)	Well ($n = 572$)	p-value
Age (years)	Median [IQR]	31.0 [14, 50]	28.0 [13, 48]	0.127
Sex	Male	297 (48.1%)	283 (49.5%)	0.667
	Female	321 (51.9%)	289 (50.5%)	
Sanitation	No	53 (8.6%)	77 (13.5%)	**0.009**
	Yes	565 (91.4%)	495 (86.5%)	
Household size	Median [IQR]	5.0 [3, 7]	5.0 [4, 7]	0.165
Socioeconomic	Low	154 (24.9%)	302 (52.8%)	**<0.001**
Status	Medium	281 (45.5%)	185 (32.3%)	
	High	183 (29.6%)	85 (14.9%)	
Education	None	100 (16.2%)	89 (15.6%)	0.591
	Primary	222 (35.9%)	201 (35.1%)	
	Secondary	210 (34.0%)	186 (32.5%)	
	Tertiary	86 (13.9%)	96 (16.8%)	

Key Findings:

- **Similarities:** No difference in Age, Sex, Household Size, or Education.
- **Differences:** Significant disparity in Socioeconomic Status ($p < 0.001$) and Sanitation ($p = 0.009$).

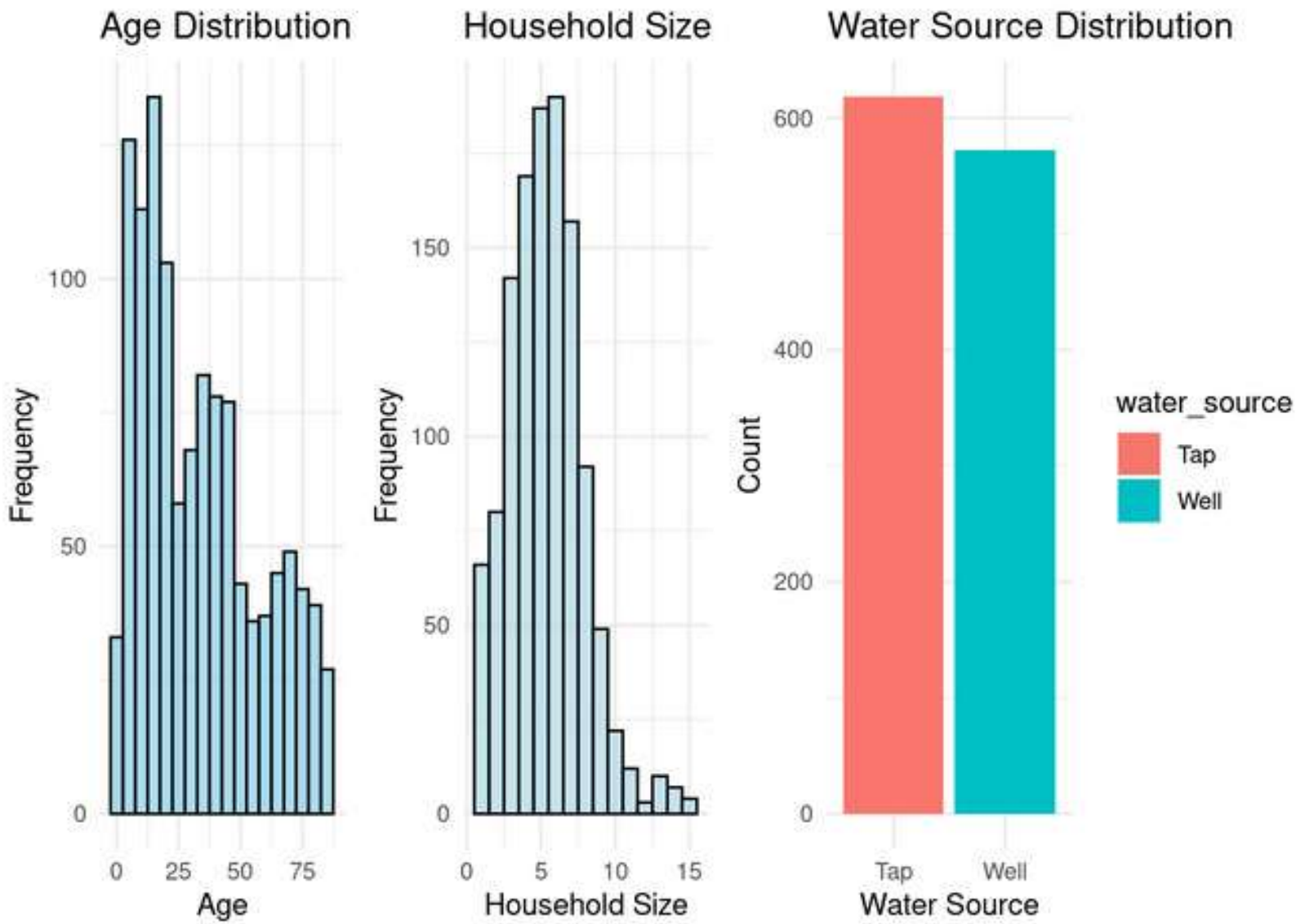

Figure 1: Decriptive Summary Plots

Follow-up and Outcome Summary

Table 2: Follow-up and Outcome Summary Stratified by Water Source

Measure	Overall	Tap Water	Well Water
Total participants	1190	618	572
Participants with ≥ 1 episode, n (%)	344 (28.9%)	92 (14.9%)	252 (44.1%)
Total diarrhoea episodes	587	169	418
Person-time at risk (person-years)	1160.46	610.26	550.20
Lost to follow-up, n (%)	53 (4.5%)	12 (1.9%)	41 (7.2%)

Key Observations:

- **High Disease Burden:** There is a striking contrast in disease occurrence, with 44.1% of well-water users experiencing diarrhoea compared to only 14.9% of tap-water users.
- **Retention & Bias:** The loss to follow-up was notably higher in the well-water group (7.2%) than the tap-water group (1.9%).
  - *This differential loss suggests a potential selection bias, as participants with adverse outcomes might have been more likely to drop out.*

Inferential Statistics: Risk and Incidence

Table 3: Measures of Association

Measure	Value
1. Incidence Risk	
Risk (Well)	0.441
Risk (Tap)	0.149
Risk Ratio (RR)	**2.96**
(95% CI)	(2.40, 3.65)
2. Incidence Rate	
Rate (Well)	759.72
Rate (Tap)	276.93
Rate Ratio (IRR)	**2.74**
(95% CI)	(2.29, 3.28)
3. Significance	
Chi-square	121.57
p-value	< 0.001

Interpretation:

1. **Risk (Probability)**
   - Well-water users are 3 times more likely to get diarrhoea than tap-water users.
   - *(RR = 2.96)*
2. **Rate (Speed)**
   - New cases occur 2.7 times faster in the well-water group.
   - *(IRR = 2.74)*

Conclusion

- Strong statistical evidence that well water increases disease risk ($p < 0.001$).

Conclusions and Limitations

1. **Conclusion**
   - Well water is a significant risk factor for diarrhoea ($p < 0.001$).
   - Users of well water face nearly 3 times the risk compared to tap water users.
   - Disease cases occur significantly faster in the well-water group.
2. **Public Health Implications**
   - Interventions (chlorination, filters) must specifically target households relying on wells.
   - Long-term investment in sanitation and tap water access is essential to reduce disease burden.
3. **Limitations**
   - Higher loss to follow-up in the well group (7.2%) may impact results.
   - Baseline differences in Wealth and Sanitation may contribute to the risk, separate from water quality.

References

World Health Organization.
Diarrhoeal disease.
Geneva: World Health Organization, 2024.
Available from: https://www.who.int/news-room/fact-sheets/detail/diarrhoeal-disease

Zhao, X., Li, Y., Wang, J., et al.
Global burden of diarrhoeal disease.
Frontiers in Public Health, 13, 2025.

United Nations Children's Fund (UNICEF) and World Health Organization.
Progress on drinking water, sanitation and hygiene: 2023 update.
New York & Geneva: UNICEF and WHO, 2023.